

Monday Warm-Up: Unit 6, Momentum I and II

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1 Memory Bank

- $\vec{p} = m\vec{v}$... Definition of momentum.
- $\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$... Momentum conservation for two objects interacting.
- $F_{\text{net}} = \Delta p / \Delta t$... Relationship between force and change in momentum.
- $F_{\text{net}} \Delta t = \Delta p$... This arrangement is known as *impulse*, the change in momentum.

2 Momentum, Impulse, and Momentum Conservation

1. Suppose a child drives a bumper car head on into the side rail, which exerts a force of 4000 N on the car for 0.200 s. (a) What impulse is imparted by this force? (b) Find the final velocity of the bumper car if its initial velocity was 2.80 m/s and the car plus driver have a mass of 200 kg. You may neglect friction between the car and floor.

2. Train cars are coupled together by being bumped into one another. Suppose two loaded train cars are moving toward one another, the first having a mass of 150,000 kg and a velocity of 0.300 m/s, and the second having a mass of 110,000 kg and a velocity of -0.120 m/s. (The minus indicates direction of motion.) What is their final velocity?

3. Two football players collide head-on in midair while trying to catch a thrown football. The first player is 95.0 kg and has an initial velocity of 6.00 m/s, while the second player is 115 kg and has an initial velocity of -3.50 m/s. What is their velocity just after impact if they cling together?

3 Final Project Construction

1. Once you have completed the above exercises, find your lab partner and discuss what system you would like to study for your final project. (a) Sketch a basic design below, labeling all important components. (b) Make a list of tools and parts you will need. (c) What subset of tools or parts will you need from our labs?